SECD 2523 - DATABASE

DATABASE CONCEPTUAL DESIGN (P2)

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1.0 INTRODUCTION

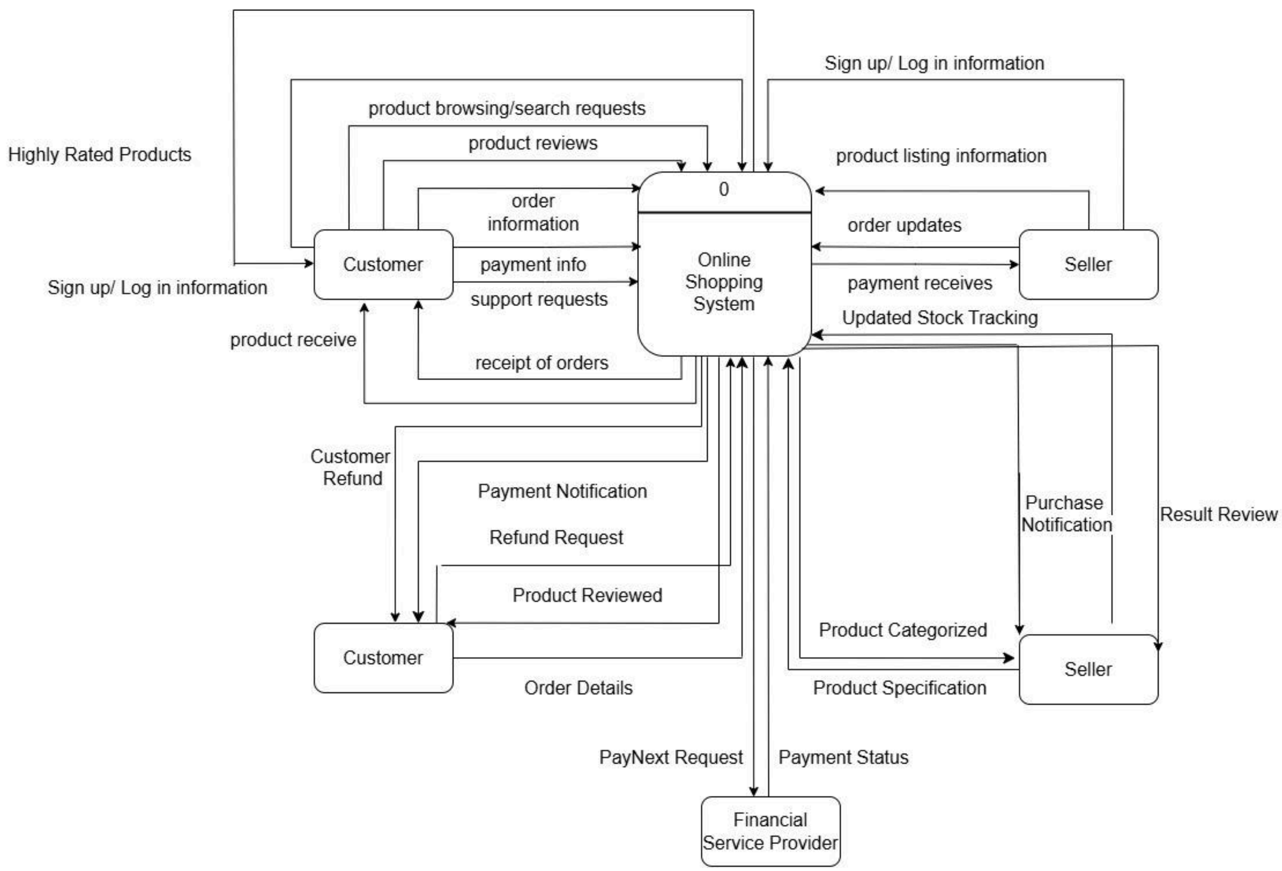
As the online shopping space further picks up in the retail world, consumers are demanding a fast, engaging, and personalized experience. Unfortunately, most applications created for this purpose, like Shopee, Lazada, and Temu, do not impress in key aspects like the intuitiveness of the user interface, personalized recommendations, efficient search, and customer engagement. These features, apart from the disconnected shopping experience, are elements of concern as far as high rates of cart abandonment, lack of personalization in customer outreach, and weak retention methods go.

The integrated principles of smart design and user-oriented design fit these needs for the Online Shopping Management System. Equipped with an advanced user interface, strong search functionality with advanced filters, an effortless checkout process, and a powerful recommendation engine, this system assures users of ease and pleasure while performing their purchase of goods. It also involves loyalty programs, sharing on social media, and a persistent shopping cart that will assure user engagement, community trust, and convenience over several sessions.

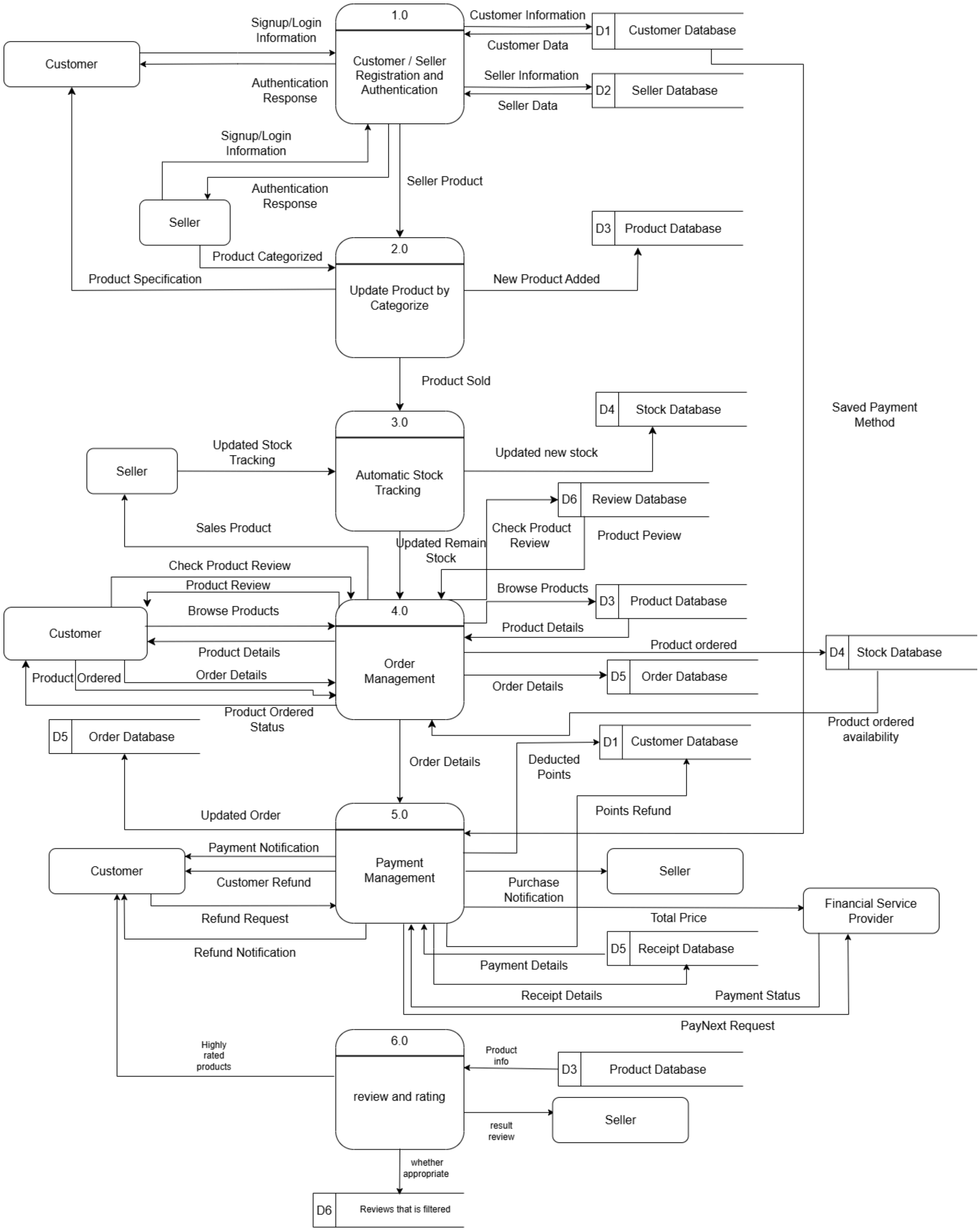
The platform offers ease of operation, thereby giving it various advantages over any competitor. Enabling a personalized experience through custom profiles and adaptive recommendations to personalized offers, encouraging repeated engagement and brand loyalty; multi-payment options integrated with an intuitive secure checkout address common pain points in online transactions, cultivating customer satisfaction and reducing abandonment. Although other platforms offer a wide range of products online, the Online Shopping Management System focuses on personalization of a customer's preference to find products and maintains user engagement. The system can meet and excel at modern consumers' demands and be a potent competitor.

2.0 DATA FLOW DIAGRAM (DFD): TO - BE

2.1 CONTEXT DIAGRAM

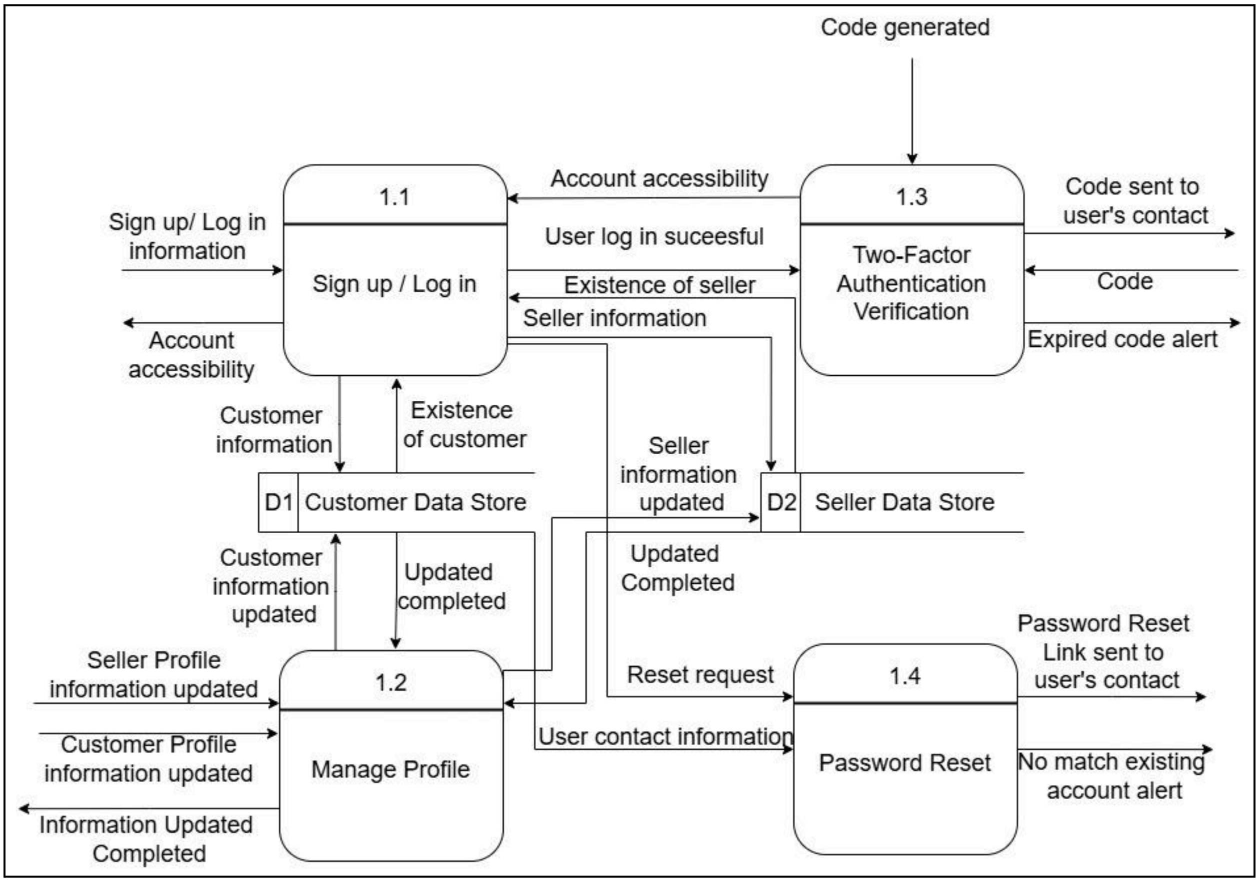


2.2 LEVEL 0 DIAGRAM

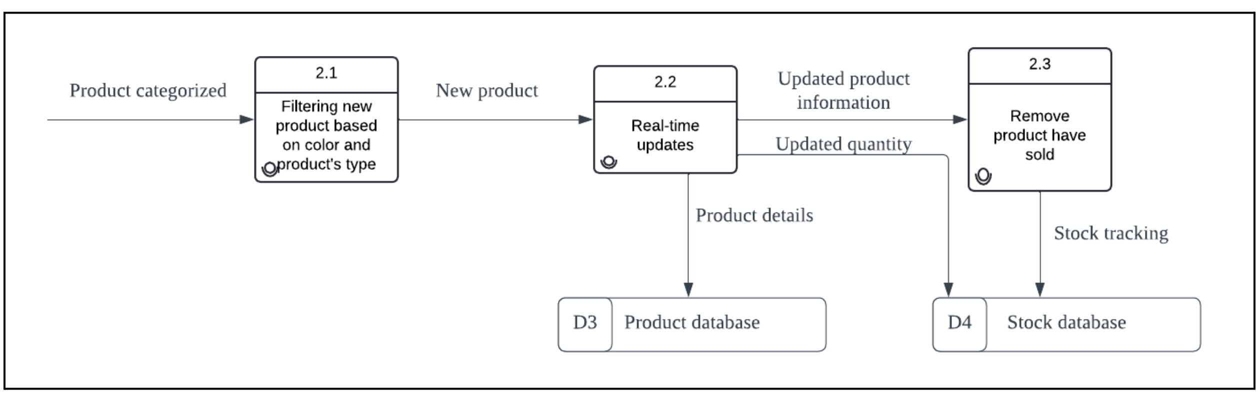


2.3 CHILD DIAGRAM

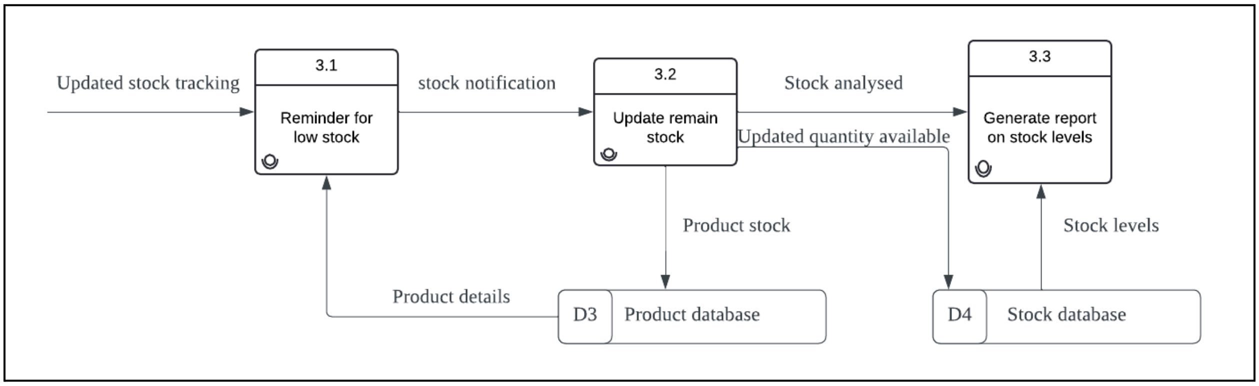
2.3.1 SIGN UP/LOGIN MANAGEMENT



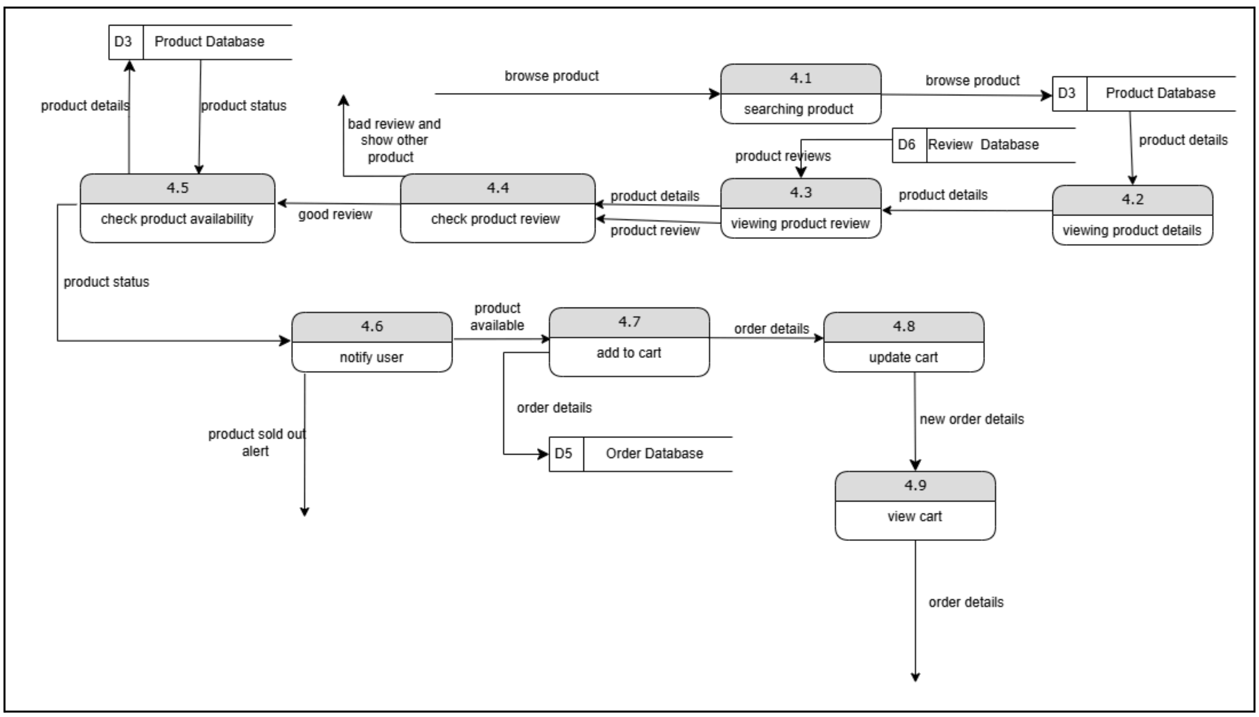
2.3.2 PRODUCT MANAGEMENT



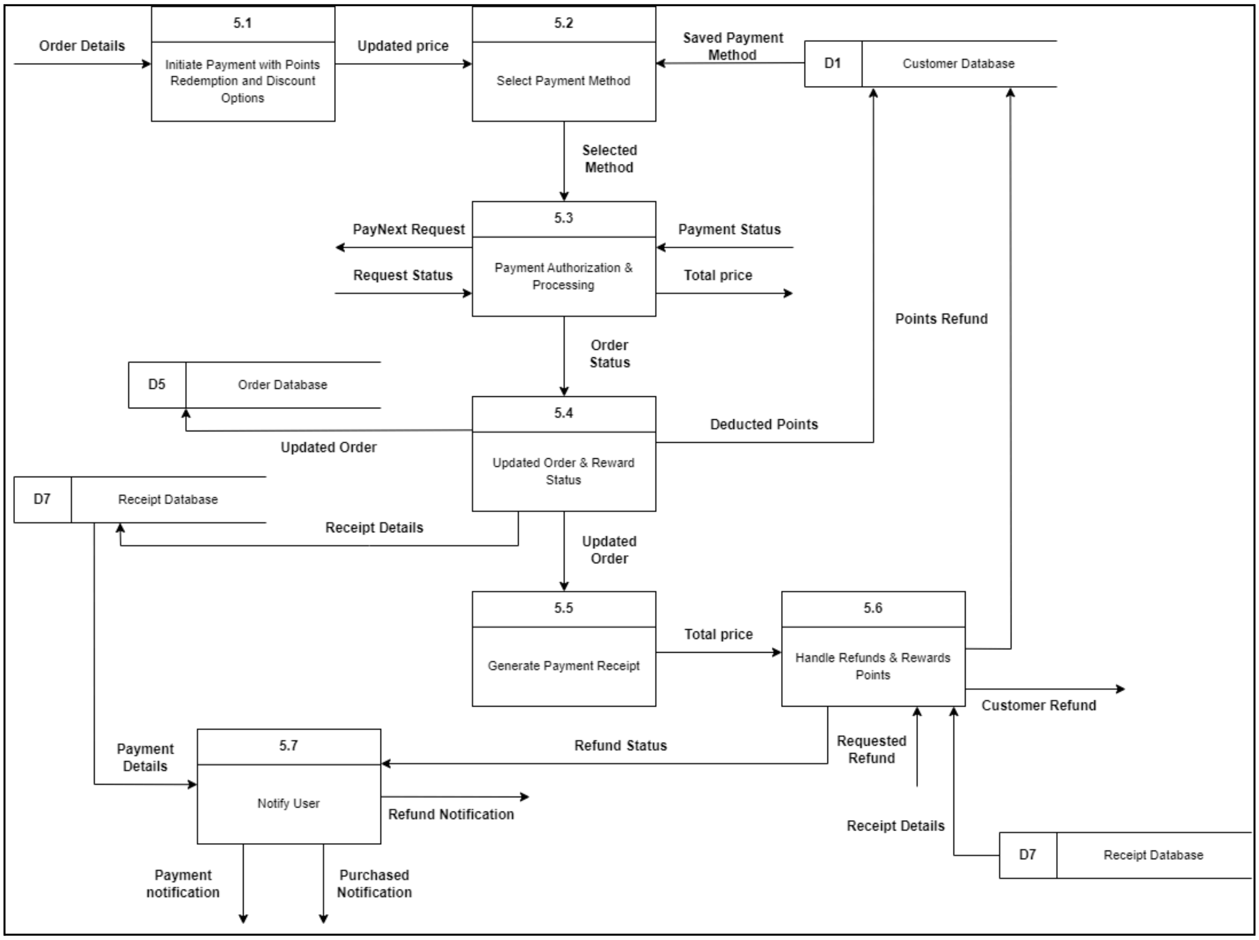
2.3.3 INVENTORY MANAGEMENT



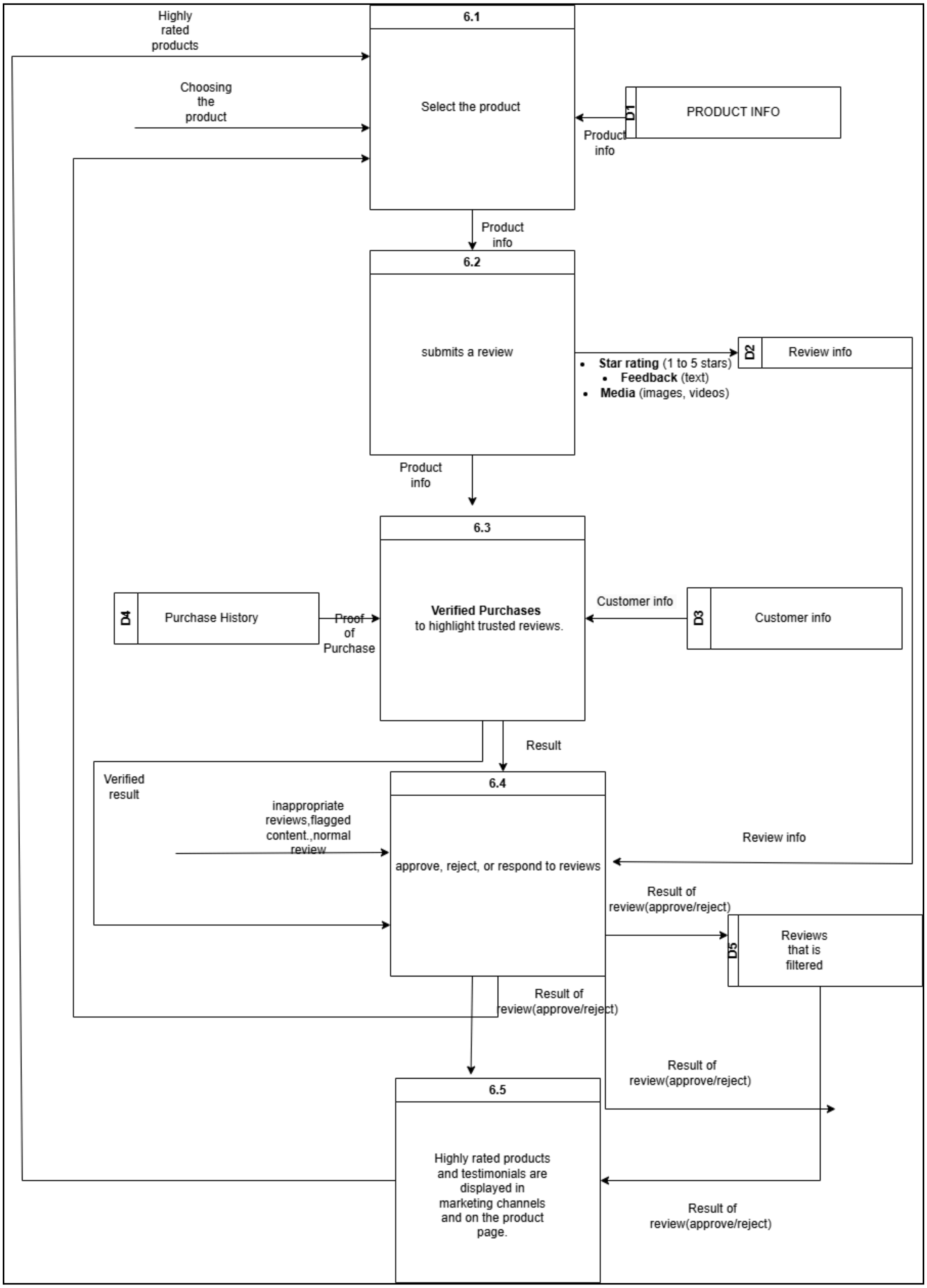
2.3.4 ORDER MANAGEMENT



2.3.5 PAYMENT MANAGEMENT



2.3.6 REVIEW AND RATING



3.0 DATA & TRANSACTION REQUIREMENT

3.1 PROPOSED BUSINESS RULE

Customer Registration:

Each customer must register an account to place orders.

A registered customer can log in to access personalized recommendations and manage their profile.

Product Management:

Each product is categorized under one or more categories.

Products are managed in the system with unique identifiers, stock quantity, and detailed descriptions.

Order Placement:

A customer can place multiple orders, each with a unique Order ID.

Orders contain one or more products and are tracked by status (e.g., pending, shipped, delivered).

Inventory Management:

Inventory levels are monitored, and products are restocked automatically when stock reaches a predefined threshold.

Each product's stock quantity is updated after an order is placed.

Payment Processing:

Payments are processed for each order, with each payment having a unique Payment ID.

Payment status (e.g., completed, pending, refunded) is updated based on transaction outcomes.

Review and Rating:

Customers can leave reviews and rate products they have purchased.

Reviews and ratings are stored and linked to specific products for customer feedback and recommendation systems.

Customer Support:

Customers can submit support tickets related to their orders or general inquiries.

Support tickets are tracked by status (e.g., open, in-progress, closed) until resolved by support staff

Loyalty and Engagement:

Loyalty programs and personalized offers are managed within the customer's profile to encourage engagement.

Customers receive points and rewards based on purchase history and engagement with the platform.

3.2 PROPOSED DATA & TRANSACTIONAL

3.2.1 PROPOSED DATA REQUIREMENT

Customer

The data stored in the Customer entity includes Customer ID, full name, contact number, email, address, and registration date. Customer ID is defined as the primary key. The name and address are composite attributes: the name consists of first and last name, while the address includes street, city, and postcode. A customer can place one or more orders and has one profile that includes preferences and purchase history.

Product

The Product entity includes Product ID, name, category, description, price, and stock quantity. Product ID is the primary key, ensuring each product entry is unique. A product is listed under one or more categories and can appear in multiple customer orders.

Order

The Order entity includes Order ID, Customer ID, date, total amount, and status (e.g., pending, shipped, delivered). Order ID is the primary key, with Customer ID used as a foreign key referencing the Customer entity. An order is placed by one customer and may contain multiple products.

Payment

The Payment entity includes Payment ID, Order ID, payment method, amount, status (e.g., completed, pending, refunded), and payment date. Payment ID is the primary key, while Order ID is a foreign key which referencing the Order entity. Each payment is associated with one order.

Inventory

The Inventory entity includes Product ID, stock level, reorder threshold, and supplier details. Product ID serves as the primary key, linking it to the Product entity. The inventory tracks each product's stock quantity and triggers restocking when the quantity reaches the reorder level.

Customer Support

Customer Support includes Support ID, Customer ID, Order ID, issue type, description, and status (e.g., open, in-progress, closed). Support ID is the primary key, with Customer ID and Order ID as foreign keys referencing Customer and Order entities. Each support entry is created for one order or general inquiry and managed by customer support staff.

3.2.2 PROPOSED TRANSACTIONAL REQUIREMENT

Data Entry

1. Enter new customer details, including name, contact information, and address for account creation.
2. Enter new product details such as name, category, description, price, and stock quantity for inventory.
3. Record new orders by logging customer details, product IDs, quantities, and order status.
4. Enter payment information for each order including method, amount, and status.
5. Record customer support requests by logging issue type, description, and status for tracking.

Data Update/Delete

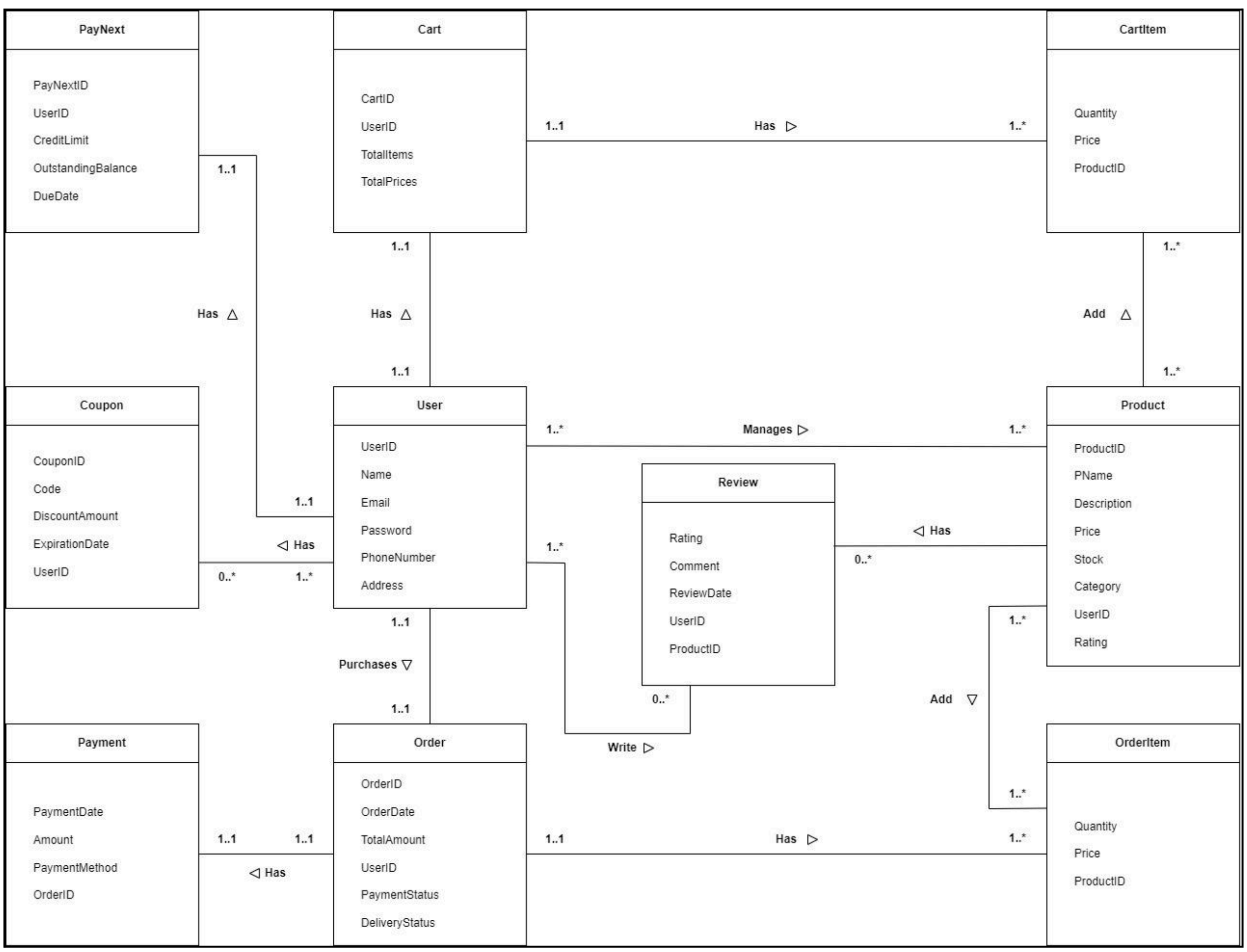
1. Update or delete customer information, such as contact number or address.
2. Update or delete product details, including price, stock quantity, or category information.
3. Modify order details, such as the status, for example, from pending to shipped.
4. Update payment status from pending to completed or refunded, as needed.
5. Edit or delete customer support requests as issues are resolved or archived.

Data Queries

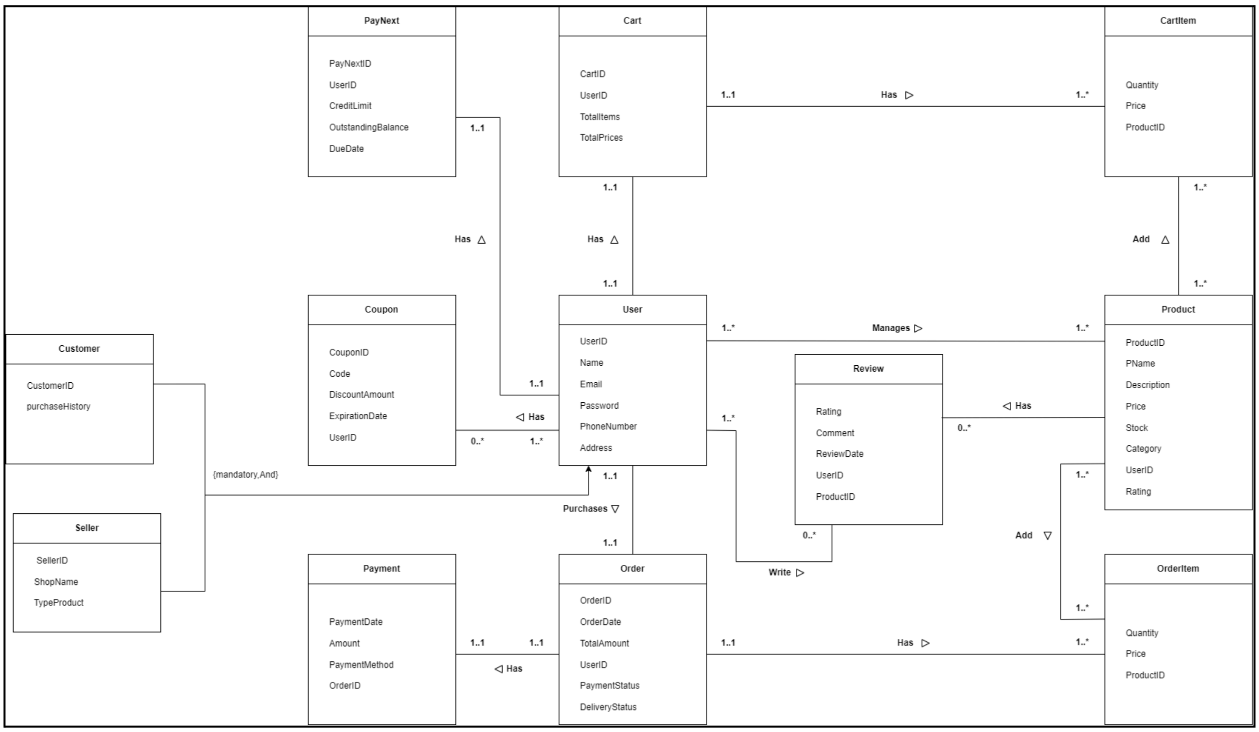
1. Display a list of all customers with their contact details and purchase history.
2. Retrieve and display the product catalog by category, with filters for price and availability.
3. Display all orders for a specific customer, including order details and statuses.
4. Query stock levels to identify products below the reorder threshold for restocking.
5. Display payment history for specific customers or orders, showing completed and pending payments.
6. Retrieve support request status and details for each order to track customer satisfaction and issue resolution.

4.0 DATABASE CONCEPTUAL DESIGN

4.1 CONCEPTUAL ERD



4.2 ENHANCED ERD (EERD)



5.0 DATA DICTIONARY

5.1 DESCRIPTION OF ENTITY

Entity	Description	Occurrence
User	User's information	User has a cart and may purchase multiple orders. They also can use coupon and write review for products.

Order	Order’s information	Order is associated with User and it can contain multiple OrderItems.It also have Payment associated with it
OrderItem	OrderItem’s information	OrderItem is part of Order and linked to product
Product	Product’s information	Product belong to multiple CartItems and be reviewed by Reviews and have multiple OrderItems associated with it.
CartItem	CartItem’s information	CartItem belong to Cart and is associated with Product
Cart	Cart’s information	Cart belongs to User and it can have multiple items in the cart
PayNext	PayNext’s information	PayNext linked to User
Coupon	Coupon’s information	Coupon can be associated with multiple User
Payment	Payment’s information	Payment linked to Order
Review	Review’s information	Review is written by User and refers to product

5.2 DESCRIPTION OF RELATIONSHIP

Entity	Multiplicity	Relationship	Multiplicity	Entity
User	1…1	Has	1…1	Cart
	1…1	Has	1…1	PayNext
	0…*	Has	0…*	Coupon
	1…*	Purchase	1…*	Order
	1…*	Manages	1…*	Product
	1…*	Write	1…*	Review
Order	1…1	Has	1…1	Payment
	1…*	Have	1…*	OrderItem
Product	1…*	Add	1…*	OrderItem
	1…*	Has	1…*	Review
	1…*	Add	1…*	CartItem
Cart	1…*	Have	1…*	CartItem

5.3 DESCRIPTION ATTRIBUTES

Entity	Attributes	Description	Data Type	Null	Multi-Valued
User	UserID	Unique identifier for each user (PK)	INT	No	No
	Name	Full name of the user.	VARCHAR	No	No
	Email	User's email address for login and contact.	VARCHAR	No	No
	Password	Password for account authentication.	VARCHAR	No	No
	PhoneNumber	User's contact phone number.	VARCHAR	No	Yes
	Address	User's residential address.	VARCHAR	Yes	No
PayNext	PayNextID	Unique identifier for each PayNext account.(PK)	INT	No	No
	UserId	User associated with this PayNext account.(FK)	INT	No	No

	CreditLimit	Maximum allowable credit for the user.	DECIMAL	No	No
	OutstandingBalance	Current outstanding balance on credit.	DECIMAL	Yes	No
	DueDate	Due date for payment of balance.	DATE	Yes	No
Cart	CartID	Unique identifier for each cart.(PK)	INT	No	No
	UserID	User associated with this cart(FK).	INT	No	No
	TotalItems	Total number of items in the cart.	INT	No	No
	TotalPrices	Total price of items in the cart.	DECIMAL	No	No
Coupon	CouponID	Unique identifier for each coupon.(PK)	INT	No	No
	Code	Code for the discount coupon.	VARCHAR	No	No

	DiscountAmount	Discount amount provided by the coupon.	DECIMAL	No	No
	UserID	User associated with the coupon(FK)	INT	Yes	No
Order	OrderID	Unique identifier for each order.(PK)	INT	No	No
	OrderDate	Date when the order was placed.	DATE	No	No
	TotalAmount	Total amount for the order.	DECIMAL	No	No
	UserID	User who placed the order.(FK)	INT	No	No
	PaymentStatus	Status of the payment (e.g., Paid, Pending).	VARCHAR	No	No
	DeliveryStatus	Status of the order delivery.	VARCHAR	No	No
OrderItem	OrderItemID	Unique identifier for each order item.(PK)	INT	No	No

	Quantity	Quantity of the product in the order item.	INT	No	No
	Price	Price per unit of the product.	DECIMAL	No	No
	ProductID	Product associated with this order item(FK)	INT	No	No
	OrderID	Order associated with this item(FK).	INT	No	No
Product	ProductID	Unique identifier for each product(PK)	INT	No	No
	PName	Name of the product.	VARCHAR	No	No
	Description	Description of the product	TEXT	Yes	No
	Price	Price of the product.	DECIMAL	No	No
	Stock	Quantity available in stock.	INT	No	No
	Category	Category the product belongs to.	VARCHAR	No	No

	SellerID	Identifier for the seller offering product(FK)	INT	No	No
	Rating	Average user rating for the product.	DECIMAL	Yes	No
Review	ReviewID	Unique identifier for each review(PK)	TEXT	No	No
	Rating	Rating given to the product by a user.	DATE	No	No
	Comment	Review text/comment by the user.	INT	Yes	No
	ReviewDate	Date when the review was submitted.	INT	No	No
	UserID	User who submitted the review.(FK)	INT	No	No
	ProductID	Product being reviewed.(FK)	INT	No	No
CartItem	CartItemID	Unique identifier for each cart item.(PK)	INT	No	No

	Quantity	Quantity of the product in the cart item.	INT	No	No
	Price	Price per unit of the product.	DECIMAL	No	No
	ProductID	Product associated with this cart item.(FK)	INT	No	No
	CartID	Cart to which this item belongs.(FK)	INT	No	No
Customer	Customer ID	Unique identifier for customer	INT	No	No
	purchaseHistory	Customer order last time	DECIMAL	No	No
Seller	ShopName	Name of seller's shop name	DECIMAL	No	No
	TypeProduct	Product that the seller is sell	DECIMAL	No	No

6.0 SUMMARY

At this stage our team has gained a thorough knowledge of the necessary groundwork for implementing an innovative Online Shopping Management System with an eye on user satisfaction and solving the major problems in the emerging e-commerce era. By developing a conceptual database design, we defined the fundamental schema to underpin the system's capabilities. This phase consisted of establishing business rules, data requirements, and data flows, as well as the design of data flow diagrams (context, level 0, and child diagrams) and a conceptual ERD.

Through the design of data flow diagrams, we elucidated the system's interaction between modules including user account management, product and inventory processing, order processing, and customer support. These diagrams clearly explain the steps of registration, searching for products, placing an order, and making a payment.

Based on the defined Data and Transaction Requirements we highlighted the requirement of a system that is able to manage personalized recommendations, user interactions, and accurate ordering. This validated the system's ability to enhance customer satisfaction and decrease cart abandonment.

The presentations of the conceptual ERD and Enhanced ERD defined fundamental database relationships. Through performing the ERD designing, we optimized entity relationships, achieving integrity and clarity of data. The data dictionary acted as a glossary, enabling the use of correct terminology. Summing up, this work provides a sufficient platform for further development of the Online Shopping Management System and moves towards the vision of providing such an efficient and friendly shopping experience.